

LAB ASSIGNMENT - 02

Exploring Packet Sniffer and Packet Analyzer:-

Question - 01: what are the different protocols you observed at the following layers of the protocol stack?

Solution:- (A) Application Layer

- HTTP
- HTTPS/TLS

(B) Transport Layer

- TCP
- UDP

(C) Network Layer

- IP

Question - 2: - what is the total amount of data being received for the following two cases?

(A) when you access `http://nevetest.com/`

(B) when you access `http://www.cornell.edu`.

Solution:-

(A) while accessing `http://nevetest.com/` (The total data received are 2274 bytes).

(B) while accessing `https://www.cornell.edu` The total data received are 16,234 bytes.

Question-03:- Answer The Following when you access the site `http://reverssl.com/`.

(A) How many HTTP GET request do you observe?
List the GET request.

Solution:- There are 3 HTTP GET requests

- GET /online HTTP/1.1
- GET /online HTTP/1.1
- GET /favicon.ico HTTP/1.1

(B) What version of HTTP the server is running

Solution:- HTTP/1.1

(C) For each of the HTTP GET request find out.

(i) The total number of TCP segment being received.

Solution:- 2 TCP segments

(ii) The total amount of data being received in the corresponding HTTP response message.

Solution:- ~~2 TCP segments~~ 2274 bytes of data

(D) Identify TCP three way handshake used to establish the connection with the `http://reverssl.com`. Specify SYN, SYN-ACK, ACK, packets and write the sequence number for each.

Solution:-

Packet
SYN
SYN-ACK
ACK

Sequence number
9179856690
1019256670 (ACK-9185666902)
9185869006 (ACK-8677859006)